

The James Webb Space Telescope (JWST) has significantly altered our understanding of Active Galactic Nuclei (AGN). These are supermassive black holes (SMBHs) which rapidly accrete matter. AGN have outsized impacts on their environments; for example, they can shut off star formation in their host galaxies and may significantly contribute to cosmic reionization.

Compared to theoretical expectations, the SMBH masses of JWST-detected AGN appear far too large for their age. Several explanations have been developed to explain this. Perhaps SMBHs can grow at super-Eddington rates. On the other hand, it may be possible that the seeds of the first black holes are larger than anticipated, including “direct-collapse” BHs. A third possibility is that the SMBH mass estimates, derived using locally-calibrated scaling relations, may be biased at high redshifts.

In addition to being too massive in an absolute sense, the AGN observed by JWST appear “overmassive” relative to their host galaxies. In the low-redshift universe, the mass of a SMBH (M_{BH}) and the total stellar mass of its host galaxy (M^*) are tightly correlated, with a ratio M_{BH}/M^* of $\sim 0.1\%$. JWST AGN appear disproportionately large, with observed M_{BH}/M^* ratios approaching or even exceeding 50%.

Intellectual Merits

I have significantly expanded the sample of JWST-detected AGNs with measured M_{BH}/M^* ratios. I analyzed a sample of broad-line AGN (BLAGN) detected using the NIRSpec/PRISM instrument on JWST as part of the CEERS and RUBIES programs. The AGN have established BH mass estimates (Taylor et al. 2024). I am deriving their stellar mass estimates by employing a novel tool called BEAGLE-AGN, which fits spectra to infer host galaxy properties. Uniquely, BEAGLE models emission from both stars and the narrow-line region (NLR) of an AGN, making these estimates more accurate.

In order to use this tool on BLAGN, I wrote a custom MCMC code to fit the lines of all sources with a 2-component gaussian model and subtract the broad component. Beyond this, emission from a BLAGN is likely to include a significant power-law continuum. In the current work, we neglect this contribution. As a result, BEAGLE must reproduce the entire observed continuum using stars alone, almost certainly making the inferred stellar masses an overestimate. Thus, any AGN which appear too large are likely to be even more overmassive than measured.

My preliminary results demonstrate that the objects in this sample of 50 AGN do appear broadly overmassive, with significantly elevated M_{BH}/M^* ratios. Moreover, I note that the M_{BH}/M^* ratios of these sources are correlated with redshift. The earlier AGN appear more overmassive, with nearby ones lying closer to the local relation. I have gathered a comprehensive sample of all JWST-detected BLAGN in the literature and uncovered the same trend. These preliminary findings may hint towards an evolutionary pathway in which SMBHs grow very rapidly in the early universe, but eventually stop as their host galaxies build up around them. I have presented these results at an international conference, will be giving a contributed talk at a national conference in May, and am preparing a manuscript on these results to be submitted to a peer-reviewed journal.

I am working with the developers of BEAGLE to actively improve the tool. We will soon include a self-consistent power-law continuum model, enabling more precise accounting of light from the AGN. This approach provides an estimate of the accretion luminosity, enabling studies into the accretion rates of these sources and BH mass estimates for narrow-line AGN. Currently, the JWST CAPERS program is taking thousands of PRISM spectra across multiple legacy fields which will uncover an unprecedented sample of BLAGN. Using this data, I will probe their broad and significant impacts on the evolution of their host galaxies and their environments at large.

Broader Impacts

As a part of my graduate studies, I have engaged in significant science outreach initiatives and mentoring. I am a lead organizer of Astronomy on Tap ATX, a platform for astronomers to give talks to public audiences. I have participated in a plethora of other outreach activities, including the UT Physics Circus and UT STEM Girl Day, to draw young learners into the scientific process. As part of the NSF-funded ISEE Professional Development Program, I am sharpening my teaching skills and will develop and deploy a novel outreach tool about telescope design at a Community Day at Stanford University. I'm also actively mentoring 8 students through the NSF-Funded Vertically Integrated Project (VIP) at UT Austin. I have guided these students through introductory materials and a preliminary project related to galaxy evolution, which they recently presented at an internal research symposium.

By leading these initiatives, I have successfully engaged students and the public in our emerging understanding of this unprecedented frontier in science.